

CITS4402 Computer Vision

Project Report

Face Detection and Matching

Preet Prasad

Student Number: 23857493

1. Introduction

This report describes the face detection and identity clustering system implemented for the CITS4402 project. The system processes 640×480 colour images (1–4 faces), detects faces, localises three landmarks (right eye, left eye, nose tip), aligns each face to a 125×125 crop, and clusters faces across a folder into unique identities. A **tkinter GUI (Task 1)** with two buttons (*Single Image*, *Bulk Processing*) wraps the pipeline; it displays the original on the left and the annotated output (boxes, landmark dots, corner thumbnails) on the right, with a status label reporting processing time, face count, and identity count. The detection-and-matching pipeline uses MediaPipe BlazeFace [2] and FaceLandmarker [6] when available, with a fully self-contained fallback built on OpenCV [3] Haar cascades [10] and the Hough circle transform [11] covered in Lab 04. For clustering, 128-D dlib ResNet embeddings [5] (via the `face_recognition` library) feed DBSCAN [4]. The table below summarises the headline measured outcomes from the naive starting point (the first working draft of each task, deliberately spec-incomplete to show what each subsequent iteration adds) to the final implementation; the sections that follow detail each task's development.

1.1 Baseline vs Final (measured on the 12 sample images)

Metric	Naive baseline	Final implementation
Face detection	11/12 images (4.jpg missed)	12/12 images, all counts correct
Landmark detection	61% both-eyes (17/28 faces)	100% both-eyes (26/26 faces)
Landmarks inside box	not measured	100% RE, 100% LE, 100% nose
Alignment success	61%, no similarity transform	100%, exact targets (0 px residual)
Identity clustering	not implemented	eps=0.38, 6 clusters from 26 faces

2. Face Detection (Task 2)

2.1 Development Iterations

Version	Change	Outcome
v1 (baseline)	Haar cascade [10] (`frontalface_default`) only, no skin filter	11/12 images; 4.jpg missed entirely
v2	Added MediaPipe BlazeFace [2] (short-range + full-range) as primary detector; Haar retained as fallback	12/12 images, but false positives: 5 boxes on 11.jpg (small body region), 1 face missed on 12.jpg (Ross failed cascade re-verification)
v3	Added skin-coverage bypass for verification (skin $\geq$ 75% AND box $\geq$ 60 px); cap result at 4 (spec maximum)	Fixed 12.jpg (Ross retained) and 11.jpg (small body box dropped); 12/12 correct
v4	Replaced YCrCb-only skin filter with three-rule majority vote (YCrCb + HSV wrap-around + normalised-RGB)	More tolerant of unusual lighting/darker skin without losing FP rejection
v5 (final)	CLAHE [12] replaces `equalizeHist` for the Haar-fallback grayscale input	12/12 maintained; better local detail for the Haar fallback path

2.2 Final Method

Primary: MediaPipe BlazeFace [2] short-range (230 KB) + full-range (1.1 MB) TFLite models, run on the original and horizontally-flipped image. Results merged with IoU-NMS (threshold 0.28, containment $>$ 72%). Each box expanded 20% up, 10% left/right, 15% down to capture the full head. Capped at 4 by ranking on confidence + skin-coverage score. Fallback when MediaPipe is unavailable: `frontalface_default` + `alt2` Haar cascades [10], 28% upward expansion, ROI re-verification, profile cascade as last resort.

2.3 Skin-Colour Segmentation (Mandatory)

Three-rule majority vote (Lab 05 technique): a pixel is skin when ≥ 2 of three rules agree — YCrCb (Cr $\in [133, 183]$, Cb $\in [75, 135]$), HSV (H $\in [0, 28] \cup [160, 179]$), normalised RGB (R $>$ G $- 5$, R $>$ B, R > 70). Morphological open + close (5×5 ellipse) before the 20% pixel-ratio threshold.

2.4 Alternatives Considered

- dlib HOG face detector: lower recall on non-frontal faces vs BlazeFace.
- OpenCV YuNet: needs a separate ONNX file not bundled with `opencv-python`.

3. Facial Landmark Detection (Task 3)

3.1 Development Iterations

Version	Change	Outcome
v1 (baseline)	`haarcascade_eye.xml` [10] only	61% both-eyes (17/28); fails on tilted, glasses, occluded faces
v2	Hough circle transform [11] with Lab 04 edge-overlap scoring on upper 58% ROI; `eye_tree_eyeglasses` and directional eye cascades as chained fallbacks; nose tip via intensity minimum of Gaussian-blurred patch	100% both-eyes on all 26 faces
v3	Added MediaPipe FaceLandmarker [6] (iris landmarks 468/473, nose landmark 1) as primary path	100% maintained; sub-pixel iris precision
v4	Per-eye position validation: reject MediaPipe landmarks where any single eye is outside the top 8–60% of the face box, or eyes differ in y by $>$ 20% face height	Caught a Chandler-in-7.jpg case where MediaPipe placed the right eye on the forehead, causing a badly tilted alignment
v5 (final)	Integral-projection profile + symmetry-mirror fallback tiers; CLAHE [12] preprocessing	100% even with MediaPipe disabled (verified by forcing the landmarker off)

3.2 Final Method

Seven-tier fallback chain (each tier only runs if the previous failed): (1) MediaPipe FaceLandmarker on a padded ROI; (2) Hough circles with Lab 04 edge-overlap scoring; (3) `haarcascade_eye_tree_eyeglasses`; (4) directional eye cascades; (5) integral-projection profile (darkest row in the upper face); (6) symmetry mirror if one eye is found; (7) proportional geometric estimate. Nose tip always from intensity minimum in the central 30–70% $\times$ 50–72% region of the face ROI.

3.3 Alternatives Considered

- dlib 68-point shape predictor: 68 MB model file, too large for submission.

4. Facial Alignment (Task 4)

4.1 Development Iterations

Version	Change	Outcome
v1 (baseline)	`getRotationMatrix2D` for rotation, then separate crop + resize. Does not use a similarity transform; landmarks not at target positions	61% success (tied to landmark failures)
v2	`cv2.estimateAffinePartial2D` with LMEDS on all three landmarks	100% success but visible tilt artefact when nose constraint is geometrically inconsistent with eyes (e.g., Monica in 7.jpg, eyes both at y=133 but face aligned with $\sim$20° rotation)
v3	Analytical 2-point similarity transform from the two eye points only	100%, no tilt; eyes exact at (40,40) and (85,40); but nose drifted from (63,70) — flagged by the lab facilitator
v4	Least-squares similarity transform over all three landmarks (closed form via `np.linalg.lstsq` on the 6-equation 4-unknown system)	100%; reduced tilt vs LMEDS but still measured max 9.5 px residual because a 4-DOF model cannot exactly fit 3 arbitrary point pairs
v5 (final)	`cv2.getAffineTransform` [3] with all three landmarks	100%; verified 0 px residual on all 26 faces — every landmark lands precisely at its target coordinate

4.2 Final Method

`cv2.getAffineTransform` takes exactly 3 source$\rightarrow$target pairs and returns the unique 6-DOF affine matrix that maps them exactly. A single `cv2.warpAffine` call with output size (125, 125) simultaneously aligns, crops, resizes, and (where needed) lightly stretches the face so that right_eye $\rightarrow (40, 40)$, left_eye $\rightarrow (85, 40)$, and nose_tip $\rightarrow (63, 70)$ are all exact. `borderMode=BORDER_REPLICATE` fills out-of-bounds pixels, preventing black borders. Landmark dots on the corner thumbnail are drawn at the fixed target coordinates and never move between images.

5. Identity Clustering (Task 5)

5.1 Development Iterations

Version	Change	Outcome
v1 (baseline)	Not implemented in the baseline code	—
v2	128-D dlib ResNet embeddings [5] via `face_recognition` (explicitly permitted by spec); DBSCAN [4] via scikit-learn [8] with `min_samples=1` and Euclidean distance	Clusters identities meaningfully; see eps sweep below
v3	Singleton-merge post-processing: merge isolated one-face clusters into the nearest compatible cluster if distance $<$ 0.515 AND no shared source image	Repairs cases where one image of a person has unusual pose/lighting
v4	Upgraded the LBP fallback descriptor from basic 256-bin 5×5 to LBP-u2 [7] 7×7 multi-scale following Ahonen et al. [1]: 59 uniform-pattern bins × 49 cells × 2 radii = 5782-D L2-normalised	LBP fallback peak silhouette [9]: 0.04 → 0.11 ($\approx 3\times$ improvement)

5.2 DBSCAN eps Sweep (final code, 26 faces)

eps	Clusters	**Silhouette** [9]
0.35	10	0.1408
0.38	**6**	**0.1021 ← selected**
0.40	4	0.0954
0.43	1	0.0000
0.45	1	0.0000

eps=0.38 was selected because 6 clusters is closest to the 5-identity ground truth (the 5 sample-image characters), with one slight over-split for a character whose appearance varies significantly between images. eps=0.35 has marginally higher silhouette (0.14) but over-splits to 10 clusters. eps=0.40 collapses two distinct identities. `min_samples=1` ensures every face joins a cluster (no noise points).

The end-to-end pipeline reports a smaller identity count than the raw DBSCAN sweep above because the v3 singleton-merge post-processing folds the six DBSCAN clusters into three identity groups: it joins isolated one-face clusters into the nearest large cluster whenever the embedding distance is $<$ 0.515 and they do not share a source image. The DBSCAN row in the *Baseline vs Final* table reports the raw 6 clusters; the GUI status bar after a *Bulk Processing* run reports the post-merge count.

5.3 Alternatives Considered

- k-means: cannot be used — number of identities is unknown a priori.
- Agglomerative clustering: rejected because DBSCAN is explicitly named in the spec.

6. Final Results

Measured on the 12 sample images (26 detected faces, all face counts match ground truth):

- Face detection: 12/12 images correct; avg detection $\approx$ 39 ms (MediaPipe path).
- Landmark detection: 100% both eyes; 100% RE, 100% LE, 100% nose inside the face box.
- Alignment: 100% success; 0 px residual on every landmark; all outputs exactly 125×125.
- Clustering: 6 DBSCAN clusters from 26 faces (eps=0.38, silhouette = 0.10).

Known limitations. (1) MediaPipe model files ($\approx$ 4.9 MB total) must accompany the script — the Haar/Hough fallback covers their absence automatically with no measurable loss on the sample set. (2) `face_recognition` (dlib) must be pip-installed for best clustering. The LBP-u2 fallback works without it but degrades the silhouette score from 0.10 to ≈ 0.07 at the same eps and over-splits more aggressively. Tasks 1–4 are unaffected by the fallback. (3) eps=0.38 over-splits one identity (6 raw DBSCAN clusters from 5 characters); eps=0.40 merges two distinct characters, so 6 is the better error. The singleton-merge post-processing then yields the 3 final identity groups reported by the GUI.

References

[1] Timo Ahonen, Abdenour Hadid, and Matti Pietikäinen. Face description with local binary patterns: Application to face recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 28(12): 2037–2041, 2006.

[2] Valentin Bazarevsky, Yury Kartynnik, Andrey Vakunov, Karthik Raveendran, and Matthias Grundmann. BlazeFace: Sub-millisecond neural face detection on mobile GPUs. *arXiv preprint arXiv:1907.05047*, 2019.

[3] Gary Bradski. The OpenCV library. *Dr. Dobb's Journal of Software Tools*, 25(11):120–125, 2000.

[4] Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu. A density-based algorithm for discovering clusters in large spatial databases with noise. In *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 226–231, 1996.

[5] Davis E. King. Dlib-ml: A machine learning toolkit. *Journal of Machine Learning Research*, 10:1755–1758, 2009.

[6] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. MediaPipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*, 2019.

[7] Timo Ojala, Matti Pietikäinen, and Topi Mäenpää. Multiresolution gray-scale and rotation invariant texture classification with local binary patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(7):971–987, 2002.

[8] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[9] Peter J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.

[10] Paul A. Viola and Michael J. Jones. Rapid object detection using a boosted cascade of simple features. In *Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR)*, volume 1, pages 511–518. IEEE, 2001. doi: $10.1109/\mathrm{CVPR}.2001.990517$.

[11] H.K. Yuen, J. Princen, J. Illingworth, and J. Kittler. Comparative study of hough transform methods for circle finding. *Image and Vision Computing*, 8(1):71–77, 1990.

[12] Karel Zuiderveld. Contrast limited adaptive histogram equalization. In Paul S. Heckbert, editor, *Graphics Gems IV*, pages 474–485. Academic Press, 1994.